

Skills Summary

Zero and Negative Exponents

Use this reference while completing your worksheet or when studying.

1. What zero and negative exponents mean

Definitions. $a^0 = 1$ for every $a \neq 0$, and $a^{-n} = \frac{1}{a^n}$ (so a^{-n} is the *reciprocal* of a^n).

Why: $\frac{a^3}{a^3} = a^{3-3} = a^0$, and a number divided by itself is 1. Keep subtracting and a^{-1} has to mean $\frac{1}{a}$ for the pattern to hold.

A negative exponent never makes a value negative. It makes it small.

Example: Evaluate 3^{-2} .

Step 1. The exponent is negative, so the definition says take the reciprocal of the positive power — nothing here is negative.

Step 2. $3^{-2} = \frac{1}{3^2}$ and $3^2 = 9$.

$$\Rightarrow \frac{1}{9}$$

Try it: Evaluate 2^{-3} .

$$\frac{8}{1} \text{ :check}$$

2. Rewriting with positive exponents

Rule. A factor with a negative exponent changes *level* and turns positive: from the numerator it moves down, from the denominator it moves up. In symbols, $\frac{1}{a^{-n}} = a^n$.

Only the factor that carries the exponent moves. In $3x^{-2}$ the 3 has no negative exponent, so it stays where it is: $3x^{-2} = \frac{3}{x^2}$, not $\frac{1}{3x^2}$.

Example: Rewrite $5x^{-3}$ with positive exponents.

Step 1. Ask which factor carries the negative exponent: only x does, so only x moves.

Step 2. Send x^{-3} to the denominator, where it becomes x^3 , and leave the coefficient 5 on top.

$$\Rightarrow \frac{5}{x^3}$$

Try it: Rewrite $\frac{2x^{-2}}{y^{-3}}$ with positive exponents.

$$\frac{x^2}{y^3} \text{ :check}$$

3. The exponent laws still work

Laws (same base). $a^m \cdot a^n = a^{m+n}$, $\frac{a^m}{a^n} = a^{m-n}$, $(a^m)^n = a^{mn}$.

These hold for negative and zero exponents too — just do the integer arithmetic on the exponents, then rewrite any negative result as a reciprocal at the very end.

Example: Simplify $x^3 \cdot x^{-7}$.

Step 1. Same base, multiplied — that is the product law, so add the exponents.

Step 2. $3 + (-7) = -4$, giving x^{-4} , which then has to be rewritten with a positive exponent.

$$\Rightarrow \frac{1}{x^4}$$

Try it: Simplify $(x^{-3})^2 \cdot x^4$.

check:
 $\frac{x^2}{1}$

4. Comparing powers

Rule. Evaluate first, compare second. Comparing the exponents instead of the values is the usual trap: 2^{-3} is bigger than 3^{-2} even though -3 is smaller than -2 .

For unit fractions, the **smaller denominator** is the larger number. And two different-looking powers can be equal.

Example: Compare 2^{-4} and 4^{-2} .

Step 1. The bases differ, so nothing can be read off the exponents — write each one as a fraction.

Step 2. $2^{-4} = \frac{1}{16}$ and $4^{-2} = \frac{1}{16}$: the same number.

$$\Rightarrow =$$

Try it: Write $>$, $<$ or $=$: 3^0 ____ 3^{-2} .

check:
 $<$